TOP, retransmits the earlier packet. This is Automatic Repeat Request Protocol (ARQ) for error control.

Flags, window and urgent fields are used to manage the TCP connection. Flags are used to connect and terminate a connection. Window field tells the transmitter the speed at which to send packets. When receiver buffer is full, the window is set to zero. This is how throughput is regulated by the transmitter sent by the receiver. Urgent field is used to send urgent information. When a receiver gets a packet with urgent field, it processes the packet with priority and acts accordingly, such as aborting transmission.

There are two variations of layer 3: user datagram protocol (UDP) and Internet control message protocol (ICMP). UDP is used when no sequence number is used. A simple example of this situation would be when a message can be put into one datagram. ICMP is simpler than UDP. Here, the message is put into one datagram only. Besides, the message is for the network and, hence, no addressing is required. ICMP is intended for TCP/IP software only.

UDP datagram is used to carry time-sensitive data like voice and video. UDP works in connection-less mode. It has fewer overhead bits, and overhead fields are quite simple. Format of UDP pack is shown in Fig. 2.

UDP headers

The 16-bit source port field indicates the address of the process at the sender, whereas process means the individual process that is in communication with the same process at the receiver.

The 16-bit destination port refers to the address of the port at the receiver that is in communication with the same process at the sender.

The 16-bit length field refers to the length of the whole UDP pack in bytes. Here, header fields are fixed, unlike in TCP. Data field is then determined from length field minus header field.

The 16-bit checksum field is used to check errors in the frame. Checksum measure is taken over the entire frame. When the receiver gets UDP frame, if calculated checksum is different from that of the transmitter, the receiver will detect the error.

A few commonly-used ports for UDP applications are 53 for DNS (domain name server), 123 for NTP (network time protocol), 161 for SNMP (simple network management protocol) and 2049 for NFS (network file system).

Above TCP/IP or UDP/IP suite is the application/higher layer. It may be layer 4 if the suite is taken as one single layer, or layer 5 if the suite is taken as two separate layers. From the Internet point of view, the suite is of single layer. Thus, layer 4 is the application layer, and this corresponds to the remaining higher layers of OSI/ISO protocol. Services of the layer include file transfer protocol (FTP), remote login, computer mail and program-to-program communication using socket programming interface and remote procedure call (RPC).

FTP allows a user, computer or terminal to get (or send) files to and from another computer. It is a tool of the Internet. Remote login is based on TELNET (network terminal protocol), which is another tool of the Internet. It allows any user to log into any computer in the network. Computer mail allows a user to send mails to another computer in the network—one classic example of this is e-mail. Each of the layers, while performing their functions, adds a header to the message/datagram at transmitting side—these headers are removed by the corresponding layer at receiving side. Layer-to-layer function is a peer process.

TCP/IP is an internetworking protocol. The Internet is an interconnected network of wide varieties of networks. Interconnection among different networks (often incompatible) is done by several methods, as given below.

- By repeaters that cover only the physical layer. An example would be an Ethernet to Ethernet connection.
- By bridges that cover physical and data link layers, corresponding to OSI/ISO protocol. An example would be connecting Ethernet LAN to token bus LAN.
- By routers that cover physical,